GLOBAL REGULARITY FOR OPTIMAL TRANSPORT FROM LIPSCHITZ DOMAINS

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Abstract. We establish global $C^{1,\alpha}$ and $W^{2,1+\varepsilon}$ estimates for quadratic optimal transport from bounded Lipschitz domains to bounded convex domains, with densities bounded above and below by positive constants. The $C^{1,\alpha}$ estimates may fail for source domains with Hölder boundary. As an application, we obtain $C^{1,\alpha}$ regularity of free boundaries in optimal partial transport up to their intersections with the fixed boundaries. The main difficulty is the failure of the usual doubling property for the source measure on sections.

1. Introduction

Let $\Omega, \Omega^* \subset \mathbb{R}^n$, $n \geq 2$, be bounded domains, and let f, g be probability densities concentrated in Ω, Ω^* respectively, satisfying

$$\lambda \leq f \leq \Lambda \quad \text{in } \Omega, \qquad \lambda \leq g \leq \Lambda \quad \text{in } \Omega^*, \tag{1.1}$$

where $0 < \lambda \leq \Lambda < \infty$. By Brenier's theorem [3], there exists a unique optimal transport map from $f\chi_\Omega\, dx$ to $g\chi_{\Omega^*}\, dy$, moreover, it is the gradient of a convex function u, with

$$(Du)_\#(f\chi_\Omega\, dx) = g\chi_{\Omega^*}\, dy. \tag{1.2}$$

When the target domain is convex, Caffarelli [5] proved strict convexity and interior $C^{1,\alpha}$ regularity of u under (1.1). If both domains are convex, global regularity follows from his boundary estimates [4, 6]. For nonconvex source domains, Andriyanova and Chen [2, Theorem 2.1] proved global C^1 regularity on Lipschitz domains under a local doubling condition, and global $C^{1,\alpha}$ regularity on convex domains with finitely many disjoint convex subsets removed. In dimension two, global C^1 regularity holds without any boundary regularity assumption on the source domain[1, 17]. Smooth perturbations of convex domains were considered in [9].

In the study of boundary regularity for optimal transport potentials, the doubling property of the corresponding Monge–Ampère measure on sections near the boundary plays a crucial role. However, the Lipschitz regularity of the source domain does not guarantee this doubling property, posing a significant difficulty. In this paper, we overcome this difficulty and establish global $C^{1,\alpha}$ regularity for arbitrary bounded Lipschitz source domains.

Theorem 1.1. *Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain and let $\Omega^* \subset \mathbb{R}^n$ be a bounded convex domain. Suppose that f, g satisfy* (1.1)*. Suppose u is a convex function satisfying* (1.2)*. Then, $u \in C^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0, 1]$.*

Remark 1.2. The standard minimal convex extension of u (see (2.2)) is differentiable on $\mathbb{R}^n$ and satisfies

$$|Du(x) - Du(z)| \leq C|x - z|^\alpha \qquad (x, z \in \mathbb{R}^n). \tag{1.3}$$

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The constants depend only on n, λ, Λ, the Lipschitz character of Ω, the diameters of the two domains, and a radius of an interior ball in Ω^*.

The Lipschitz assumption cannot be replaced by an arbitrary Hölder boundary condition. For every $n \geq 3$, in Proposition 3.3 we construct a bounded connected $C^{0,\beta}$ source domain, with $\beta = (n-2)/(2(n-1))$, and a bounded convex target domain, with both densities equal to one, for which the potential function u is not C^1 near the boundary.

As an application of Theorem 1.1, we obtain a global Sobolev estimate. Interior $W^{2,1}$ regularity was proved by De Philippis and Figalli [14], and the improvement to $W^{2,1+\varepsilon}$ was obtained by De Philippis, Figalli and Savin [15], and independently by Schmidt [27]. Chen, Liu and Wang [10] established global $W^{2,1+\varepsilon}$ regularity for bounded convex domains in dimension two and for bounded convex domains with $C^{1,1}$ boundaries in higher dimensions. Savin and Yu [26] subsequently extended this estimate to arbitrary bounded convex domains in all dimensions. For nonconvex planar sources, Mooney and Rakshit [24] obtained sharp estimates for constant densities on convex domains with finitely many, separated uniformly convex holes removed, and Hu and Li [19] established global $W^{2,1+\varepsilon}$ regularity for polygonal sources. The following result allows an arbitrary bounded Lipschitz source domain.

Theorem 1.3. *Under the assumptions of Theorem 1.1, there are $\varepsilon > 0$ and $C < \infty$, depending only on the same data, such that $u \in W^{2,1+\varepsilon}(\Omega)$ and*

$$\int_\Omega |D^2 u|^{1+\varepsilon}\, dx \leq C. \tag{1.4}$$

For convex domains, stronger estimates hold under continuity assumptions on the densities. Chen, Liu and Wang [11] proved global $C^{2,\alpha}$ and $W^{2,p}$ estimates for $C^{1,1}$ convex domains. Collins and Tong [13] established global $W^{2,p}$ regularity on arbitrary bounded convex domains for positive Dini continuous densities, and global $C^{2,\alpha}$ regularity on bounded convex $C^{1,\alpha}$ domains for positive C^α densities. Chen, Li and Wang [8] treated positive continuous densities.

A second application concerns free boundaries in optimal partial transport. For separated strictly convex supports, Caffarelli and McCann [7] proved continuity of the transport map up to the boundary and identified the transport direction with the free-boundary normal. Chen, Liu and Wang [12] proved that the active region is a Lipschitz domain and obtained global $C^{1,\alpha}$ regularity of the free boundary when the entire target mass is transported.

When the transported mass is strictly less than both the source and target masses, however, an intersection point of the source free and fixed boundaries may be mapped to an intersection point of the target free and fixed boundaries. Near such points, the active regions need not be convex, and the corresponding Monge–Ampère measures need not satisfy the doubling property, which creates a serious difficulty. In this paper, we combine the method used to prove Theorem 1.1 with the argument for Lipschitz regularity of the active regions developed in [12] to obtain the same free-boundary regularity when both measures have inactive parts.

Corollary 1.4. *Let $\Omega, \Omega^* \subset \mathbb{R}^n$ be bounded strictly convex domains with disjoint closures, and let f, g satisfy* (1.1)*. Fix*

$$0 < m < \min\left\{\int_\Omega f\, dx, \int_{\Omega^*} g\, dy\right\}. \tag{1.5}$$

Let $U \subset \Omega$ and $V \subset \Omega^*$ be the active regions for quadratic optimal partial transport of mass m, with optimal map Du, and set

$$\mathcal{F} := \partial U \cap \Omega, \qquad \mathcal{F}^* := \partial V \cap \Omega^*. \tag{1.6}$$

Then, $u \in C^{1,\alpha}(\overline{U})$, and $\overline{\mathcal{F}}$, $\overline{\mathcal{F}^*} \in C^{1,\alpha}$, for some $\alpha \in (0,1)$.

Remark 1.5. The above result also holds for general bounded convex domains. Removing the strict convexity assumption requires only technical modifications, which we omit here.

The main difficulty in proving Theorem 1.1 is that the source measure need not be doubling on sections near $\partial\Omega$, so the usual section estimates do not directly imply the required decay of section diameters. We overcome this difficulty by establishing an enlargement property through an inward affine deformation and the Monge–Ampère comparison principle, without relying on a doubling estimate. By iterating this property, we obtain a power separation estimate for the dual potential v, which implies global $C^{1,\alpha}$ regularity of u by Legendre duality.

The Sobolev estimate follows by combining the interior estimate of [15, Theorem 1.1] and the estimate for the number of sections in a Vitali covering in [22, proof of Lemma 3.5] with the Hölder estimate for Du. The latter controls the Hessian mass near $\partial\Omega$. For partial transport, we first establish Lipschitz regularity of the active regions by combining the argument of [12] with [20, Proposition 4.4]. Adding the inactive target mass, as in [7], gives a transport with convex target. The quadratic obstacle relation ensures that the localization point lies in the closure of the active source region, allowing us to adapt the proof of Theorem 1.1 and obtain global Hölder continuity of the transport map. The formula for the unit normal then implies that the free boundaries are $C^{1,\alpha}$ up to the fixed boundaries.

The remaining of this paper is organised as follows. In section 2 we collect the notation and standard facts used below. In section 3, we prove Theorem 1.1 and construct the counterexample. In ections 4 and 5 we prove Theorem 1.3 and Corollary 1.4, respectively.

2. Preliminaries

Let Ω be a bounded Lipschitz domain and let Ω^* be a bounded convex domain. Let f, g be probability densities satisfying (1.1), and let u be a convex potential satisfying (1.2). Let v be a convex potential for the optimal transport from (Ω^*, g) to (Ω, f), so that

$$(Dv)_\sharp(g\chi_{\Omega^*}\,dy) = f\chi_\Omega\,dx. \tag{2.1}$$

We extend both potentials to $\mathbb{R}^n$ by their standard minimal convex extensions. Namely, we set

$$\tilde{u}(x) := \sup\{\ell(x) : \ell \text{ is affine, } \ell \le u \text{ in } \Omega,\ D\ell \in \Omega^*\}, \tag{2.2}$$

for every $x \in \mathbb{R}^n$, and define

$$\tilde{v}(x) := \sup\{\ell(x) : \ell \text{ is affine, } \ell \le v \text{ in } \Omega^*,\ D\ell \in \Omega\}. \tag{2.3}$$

Both extensions are convex and globally Lipschitz. In the following, unless specifically mentioned, u, v denote these extensions. The two potentials are Legendre dual on the transport domains. Thus, for $y \in \Omega^*$,

$$v(y) = \sup_{x\in\Omega}\{x \cdot y - u(x)\}. \tag{2.4}$$

Equivalently, v is the Legendre transform of u restricted to Ω.

Let $\mathcal{O}$ be an open set, and let $\phi : \mathcal{O} \to \mathbb{R}$ be a convex function. The subgradient of ϕ at $x \in \mathcal{O}$ is

$$\partial\phi(x) := \{p : \phi(z) \geq \phi(x) + p \cdot (z - x) \text{ for all } z \in \mathcal{O}\}.$$

The Monge-Ampère measure is defined as

$$\mu_\phi(E) := |\partial\phi(E)|, \quad \text{where } \partial\phi(E) := \bigcup_{x\in E} \partial\phi(x),$$

for any Borel set $E \subset \mathcal{O}$.

By the definitions of the extensions,

$$u(x) = \max_{p\in\overline{\Omega^*}}\{x \cdot p - v(p)\}, \qquad \partial u(\mathbb{R}^n) \subset \overline{\Omega^*}.$$

In particular,

$$y \in \partial u(x) \quad \Longleftrightarrow \quad u(x) + v(y) = x \cdot y \qquad (y \in \overline{\Omega^*}). \tag{2.5}$$

Since Ω^* is convex, Caffarelli [5] showed that the extended transport potential u satisfies the Monge–Ampère inequality in the Alexandrov sense. In particular,

$$\frac{\lambda}{\Lambda}\chi_\Omega\, dx \leq \mu_u \leq \frac{\Lambda}{\lambda}\chi_\Omega\, dx. \tag{2.6}$$

Caffarelli also proved the strict convexity and local $C^{1,\alpha}$ regularity of u in Ω.

A constant is called universal if it depends only on the data in Theorem 1.1. We denote positive universal constants by c, C, whose values may change from line to line. The Lipschitz character of Ω comprises the radii, Lipschitz bounds and number of charts in a finite boundary covering. After translating the source, we have $\sup_\Omega |x| + \mathrm{Lip}(v) \leq C$.

The following elementary property will be useful to us. For every $p \in \overline{\Omega^*}$, there exists an x satisfying

$$x \in \overline{\Omega} \cap \partial v(p), \qquad u(x) + v(p) = x \cdot p. \tag{2.7}$$

Indeed, by (1.2) we can choose differentiability points $p_j \in \Omega^*$ approaching p with $Dv(p_j) \in \Omega$. Passing to a subsequence, $Dv(p_j) \to x \in \overline{\Omega}$.The desired property (2.7) follows.

We refer to [16, 18] for more background on Monge–Ampère measures and sections. The following basic comparison principle for the Monge–Ampère equation is important for Proposition 3.1.

Lemma 2.1. *If $\mathcal{O}$ is bounded and open, and ϕ, ψ are finite convex functions in an open convex neighbourhood of $\overline{\mathcal{O}}$, then*

$$\left.\begin{array}{ll} \psi \leq \phi & \text{on } \partial\mathcal{O}, \\ \mu_\psi \geq \mu_\phi & \text{in } \mathcal{O} \end{array}\right\} \quad \Longrightarrow \quad \psi \leq \phi \text{ in } \mathcal{O}. \tag{2.8}$$

We also recall the standard volume estimate for sections [16, Lemma 4.6]. Let φ be strictly convex and satisfy $\lambda\, dx \leq \mu_\varphi \leq \Lambda\, dx$ in an open set U. For $x \in U$ and $h > 0$, let

$$S_h(x) := \{y \in U : \varphi(y) < \varphi(x) + D\varphi(x) \cdot (y - x) + h\}.$$

Whenever $S_h(x) \Subset U$, we have

$$ch^{n/2} \leq |S_h(x)| \leq Ch^{n/2}, \tag{2.9}$$

where $c, C > 0$ depend only on n, λ, Λ.

3. Global Hölder regularity

The main point is a quantitative strict convexity estimate for the dual potential v. If v is close to an affine function on a segment, we show that it remains close to another affine function on a larger convex set. The error increases by a fixed factor at each step. A packing argument then implies a quantitative strict convexity estimate for v, from which the Hölder bound follows by Legendre duality.

Proposition 3.1. *There are constants $\varepsilon_*, c_* > 0$ and $K_* \geq 2$, depending only on the data in Theorem 1.1, with the following property. Let $A \subset \overline{\Omega^*}$ be compact and convex, with $\delta = \operatorname{diam} A > 0$, and let $\mathcal{L}$ be an affine function satisfying $\mathcal{L} \leq v$ on $\overline{\Omega^*}$. Suppose that*

$$\sup_A (v - \mathcal{L}) \leq \varepsilon, \qquad 0 < \varepsilon \leq \varepsilon_* \delta.$$

Then there are $p_+ \in \overline{\Omega^*}$ and an affine function $\mathcal{L}_+$ satisfying $\mathcal{L}_+ \leq v$ on $\overline{\Omega^*}$ such that

$$\operatorname{dist}(p_+, A) \geq c_* \delta \quad \text{and} \quad \sup_{\operatorname{conv}(A \cup \{p_+\})} (v - \mathcal{L}_+) \leq K_* \varepsilon. \tag{3.1}$$

3.1. **Hölder continuity.** We first explain why Proposition 3.1 implies the theorem. Within a fixed diameter range, only a bounded number of points can be added at the stated distance from the preceding sets. Thus each doubling of the diameter costs at most a fixed factor in the affine error. Since the target is bounded, the initial error must have a power lower bound.

Lemma 3.2. *There are constants $c > 0$ and $\sigma \geq 1$, depending only on the data in Theorem 1.1, such that, for every compact convex set $A \subset \overline{\Omega^*}$ with $\delta = \operatorname{diam} A > 0$ and every affine function $\mathcal{L}$ satisfying $\mathcal{L} \leq v$ on $\overline{\Omega^*}$,*

$$\sup_A (v - \mathcal{L}) \geq c \delta^{\sigma+1}. \tag{3.2}$$

Proof. Fix $\varepsilon > \sup_A (v - \mathcal{L})$. Let $A_0 = A$ and $\mathcal{L}_0 = \mathcal{L}$. We apply Proposition 3.1 repeatedly. At the jth step, $\mathcal{L}_j$ is the supporting affine function obtained at the previous step and A_j is the corresponding section, with $\delta_j = \operatorname{diam} A_j$. As long as $K_*^j \varepsilon \leq \varepsilon_* \delta_j$, Proposition 3.1 applies to $(A_j, \mathcal{L}_j)$ and obtains a new point p_{j+1} and a supporting affine function $\mathcal{L}_{j+1}$ such that the enlarged section A_{j+1} contains A_j and

$$\operatorname{dist}(p_{j+1}, A_j) \geq c_* \delta_j,$$

where K_*, ε_*, c_* are the constants in Proposition 3.1.

First, we estimate $m_k := \#\{j : \delta_j \in [2^k \delta, 2^{k+1}\delta)\}$ with $\delta = \operatorname{diam} A$. Suppose

$$\{j_k, j_k + 1, \cdots, j_k + m_k - 1\} = \{j : \delta_j \in [2^k \delta, 2^{k+1}\delta)\}.$$

Denote $d_k = 2^k \delta$. Fix a point $q_0 \in A_{j_k}$. By the construction, we can find $q_1 \in A_{j_k+1}$, such that $\operatorname{dist}(q_1, A_{j_k}) \geq c_* \delta_{j_k} \geq c_* d_k$. Repeat this process, we can find

$$q_i \in A_{j_k+i} \ (0 \leq i \leq m_k - 1) \ \ s.t. \ \ \operatorname{dist}(q_i, q_l) \geq c_* d_k \ (0 \leq i < l \leq m_k - 1).$$

Since $\operatorname{diam} A_{j_k+m_k-1} < 2^{k+1}\delta = 2d_k$, we have $|q_i - q_0| < 2d_k$ for all $0 \leq i \leq m_k - 1$. Hence, $B_{\frac{1}{2}c_* d_k}(q_i)$, for $i = 0, 1, \cdots, m_k - 1$, are disjoint and

$$\bigcup_{i=0}^{m_k - 1} B_{\frac{1}{2}c_* d_k}(q_i) \subset B_{2d_k + \frac{1}{2}c_* d_k}(q_0).$$

By comparing the volume, we find that m_k has an upper bound, independent of k,

$$m_k \leq \frac{(2 + \frac{1}{2}c_*)^n}{(\frac{1}{2}c_*)^n} := N. \tag{3.3}$$

Since all the sets A_j lie in $\overline{\Omega^*}$, their diameters are bounded by $R_* := \operatorname{diam}\Omega^*$. If $m_k > 0$, then $2^k\delta \le R_*$, and therefore

$$0 \le k \le \left\lfloor \log_2 \frac{R_*}{\delta} \right\rfloor.$$

Thus there are at most $1 + \lfloor \log_2(R_*/\delta) \rfloor$ indices k for which $m_k > 0$. By (3.3), the total number of steps is at most

$$\bar{N} = N\left(1 + \left\lfloor \log_2 \frac{R_*}{\delta} \right\rfloor\right). \tag{3.4}$$

The iteration must stop at some $m \le \bar{N}$, where $K_*^m \varepsilon > \varepsilon_* \delta_m \ge \varepsilon_* \delta$. Consequently,

$$\varepsilon > \varepsilon_* \delta K_*^{-\bar{N}} \ge \varepsilon_* K_*^{-N} R_*^{-\sigma} \delta^{\sigma+1},$$

where we take $\sigma = N \log_2 K_* \ge 1$ and use (3.4).

The estimate (3.2) follows by taking $c = \varepsilon_* K_*^{-N} R_*^{-\sigma}$ and letting $\varepsilon \to \sup_A(v - \mathcal{L})$. □

Proof of Theorem 1.1. Let $x^{(0)}, x^{(1)} \in \mathbb{R}^n$, and choose $y^{(0)} \in \partial u(x^{(0)})$ and $y^{(1)} \in \partial u(x^{(1)})$. The affine function

$$\mathcal{L}_{x^{(0)}}(z) := x^{(0)} \cdot z - u(x^{(0)})$$

satisfies $\mathcal{L}_{x^{(0)}} \le v$ on $\overline{\Omega^*}$, with equality at $y^{(0)}$. By duality and convexity, we have

$$\begin{aligned} 0 \le v(y^{(1)}) - \mathcal{L}_{x^{(0)}}(y^{(1)}) &= u(x^{(0)}) - u(x^{(1)}) + (x^{(1)} - x^{(0)}) \cdot y^{(1)} \\ &\le (x^{(1)} - x^{(0)}) \cdot (y^{(1)} - y^{(0)}) \\ &\le |x^{(1)} - x^{(0)}|\, |y^{(1)} - y^{(0)}|. \end{aligned}$$

For $y^{(1)} \ne y^{(0)}$, by applying (3.2) to this segment, we have

$$c|y^{(1)} - y^{(0)}|^{\sigma+1} \le |y^{(1)} - y^{(0)}|\, |x^{(1)} - x^{(0)}|.$$

Taking $x^{(1)} = x^{(0)}$, it follows that every subdifferential is a singleton. Hence u is differentiable everywhere, and

$$|Du(x^{(1)}) - Du(x^{(0)})| \le c^{-1/\sigma} |x^{(1)} - x^{(0)}|^{1/\sigma},$$

which is exactly (1.3) with $\alpha = 1/\sigma$. □

3.2. **Proof of Proposition 3.1.** We first construct a small section of u around a point of $\overline{\Omega}$. An inward affine deformation and the Monge–Ampère comparison principle yield a short segment along which u increases sufficiently fast. By duality, this gives a target point separated from A. The short source displacement also controls the increase of v above a suitable affine lower bound along a segment joining this point to a point of A. A convex combination along this segment then gives the required separation and affine error bound.

Proof. Note that the constants c_*, K_*, ε_* will be chosen below in terms of the data. We begin by normalizing A so that it contains a symmetric chord of length comparable to δ. This chord will determine the direction in which the source section is thin. By John's lemma [21] in the affine hull of A, choose $p_0 \in A$ such that $-(A - p_0)/n \subset A - p_0$. Make the paired changes

$$\tilde{v}(q) := v(p_0 + q) - v(p_0), \qquad \tilde{u}(x) := u(x) - x \cdot p_0 + v(p_0),$$

translate A, Ω^* and $\mathcal{L}$ accordingly, and omit the tildes. These changes preserve the affine errors, the measure bounds (2.6), and δ. By duality and (2.7),

$$v(0) = 0, \qquad u \ge 0, \qquad \min_{\overline{\Omega}} u = 0, \qquad -A/n \subset A.$$

Choosing a farthest point of A from zero and dividing it by n, we obtain p_1 such that

$$[-p_1, p_1] \subset A, \qquad \frac{\delta}{2n} \le |p_1| \le \delta. \tag{3.5}$$

Step 1. Selection of $x_0 \in \overline{\Omega}$. We first select a point $x_0 \in \overline{\Omega}$ where u is *bounded by a fixed multiple of ε* and *admits a quadratic lower bound.* This lower bound will localize the section in Step 2. We choose x_0 by minimizing an exponentially weighted function, and use duality to show that it lies in $\overline{\Omega}$.

Fix $M_0 > 1$ with $\overline{\Omega} \subset B_{M_0-1}$, an interior ball $B_c(q_0) \subset \Omega^*$, and $R_0 = 2M_0/c + 1$. Then $q_0 + x/R_0 \in B_{c/2}(q_0)$ for $x \in \overline{B}_{M_0}$. Choose a uniform radius $r \in (0,1)$ for the inward translations of a finite Lipschitz atlas of Ω, small enough that

$$r(M_0 + R_0 \operatorname{diam} \Omega^*) \le \frac{1}{2}. \tag{3.6}$$

Choose $x_0 \in \overline{B}_{M_0}$ to minimize

$$(u(x) + \varepsilon) \exp\left(-\frac{4(n+1)}{r^2}|x + R_0 q_0|^2\right) \qquad (x \in \overline{B}_{M_0}).$$

Such a minimizer exists by continuity and compactness. By minimality,

$$u(x) + \varepsilon \ge (u(x_0) + \varepsilon) \exp\left(\frac{4(n+1)}{r^2}\Big(|x + R_0 q_0|^2 - |x_0 + R_0 q_0|^2\Big)\right) \qquad (x \in \overline{B}_{M_0}). \tag{3.7}$$

Let $z \in \overline{\Omega}$ satisfy $u(z) = 0$. It then follows by applying (3.7) at z that

$$\varepsilon \le u(x_0) + \varepsilon \le \varepsilon \exp\left(\frac{4(n+1)}{r^2}\Big(|x_0 + R_0 q_0|^2 - |z + R_0 q_0|^2\Big)\right) \le C\varepsilon, \tag{3.8}$$

where C depends only on M_0, R_0, $\operatorname{diam} \Omega^*$ and r, since $|x_0 + R_0 q_0| \le M_0 + R_0 \operatorname{diam} \Omega^*$.

Let $p := \kappa(x_0 + R_0 q_0)$, where

$$\kappa := \frac{8(n+1)}{r^2}(u(x_0) + \varepsilon).$$

Note that if x_0 is in the interior of $\overline{B}_{M_0}$, by (3.7), we have $p \in \partial u(x_0)$. By (3.8), fix a constant $C_0 \ge 1$, depending only on the data in Theorem 1.1, such that

$$\kappa \le C_0 \varepsilon. \tag{3.9}$$

By (3.7) and using $e^t \ge 1 + t$, we obtain

$$u(x_0 + X) \ge u(x_0) + p \cdot X + \frac{\kappa}{2}|X|^2 \qquad (x_0 + X \in \overline{B}_{M_0}). \tag{3.10}$$

Since $q_0 + x_0/R_0 \in B_{c/2}(q_0)$ and $0 \in \overline{\Omega^*}$, convexity of Ω^* implies

$$p = \kappa R_0(q_0 + x_0/R_0) \in \Omega^* \quad \text{whenever} \quad \kappa R_0 \le 1/2,$$

as ensured by (3.9) and the final choice of ε_*. Take $y \in \overline{\Omega} \cap \partial v(p)$. Then, by (2.7),

$$\begin{aligned} u(y) &= y \cdot p - v(p) \\ &\le y \cdot p + u(x_0) - x_0 \cdot p \\ &= u(x_0) + p \cdot (y - x_0). \end{aligned}$$

Combining this with (3.10) at $X = y - x_0$ we have

$$\frac{\kappa}{2}|y - x_0|^2 \le 0.$$

Hence $y = x_0$, so $x_0 \in \overline{\Omega}$. We have

$$x_0 \in \overline{\Omega}, \qquad p \in \partial u(x_0), \qquad 0 \le u(x_0) \le C\varepsilon. \tag{3.11}$$

Step 2. A localized section. Let

$$h := 2(n+1)(u(x_0)+\varepsilon), \qquad \tilde{\Omega} := \Omega - x_0,$$

and normalize the potentials by

$$\tilde{u}(X) := u(x_0+X) - u(x_0) + h,$$
$$\tilde{v}(q) := v(q) - x_0 \cdot q + u(x_0) - h.$$

These changes preserve duality, the measure bounds (2.6). We have $\tilde{u}(0) = h$ and

$$\kappa = \frac{4h}{r^2}. \tag{3.12}$$

Define

$$D := \{X : \tilde{u}(X) < 2h + p \cdot X\}.$$

Then $0 \in D$.

Claim. The section D satisfies

$$\begin{aligned} &\overline{D} \subset \overline{B}_{r/\sqrt{2}}, \\ &\tilde{u}(X) < \kappa, \qquad |p_1 \cdot X| < \kappa \quad (X \in \overline{D}). \end{aligned} \tag{3.13}$$

Proof of the claim. For $q \in A$, by duality we have

$$v(-q/n) + u(x_0) \ge (-q/n) \cdot x_0.$$

Combining this with $-q/n \in A$, we have

$$\begin{aligned} 0 \le v(q) - x_0 \cdot q + u(x_0) &\le v(q) + nv(-q/n) + (n+1)u(x_0) \\ &\le (n+1)(\varepsilon + u(x_0)) = h/2, \end{aligned} \tag{3.14}$$

where the last inequality follows from $v \le \mathcal{L} + \varepsilon$ on A and $\mathcal{L}(q) + n\mathcal{L}(-q/n) = (n+1)\mathcal{L}(0) \le 0$. Thus

$$-h \le \tilde{v} \le -h/2 \quad \text{on } A. \tag{3.15}$$

For $X \in \overline{D} \cap \overline{B}_r$, (3.10) and the definition of D give

$$\frac{\kappa}{2}|X|^2 \le \tilde{u}(X) - h - p \cdot X \le h.$$

Thus $|X|^2 \le 2h/\kappa = r^2/2$ on $\overline{D} \cap \overline{B}_r$. By convexity and $0 \in D$, this implies $\overline{D} \subset \overline{B}_{r/\sqrt{2}}$.

By the definition of p and (3.6), $r|p| \le \kappa/2$. Hence, on $\overline{D}$,

$$\tilde{u}(X) \le 2h + r|p| \le \kappa\left(\frac{r^2}{2} + \frac{1}{2}\right) < \kappa.$$

Finally, since $\pm p_1 \in A$ and $\tilde{v} \le -h/2$ on A, duality gives

$$\tilde{u}(X) \ge |p_1 \cdot X| + h/2 \qquad (X \in \mathbb{R}^n). \tag{3.16}$$

Thus $|p_1 \cdot X| < \kappa$ on $\overline{D}$, proving the claim.

Step 3. Construction of the comparison functions. Recall that p_1 satisfies (3.5). After reversing p_1 if necessary, choose a vector η and a uniform constant $s_0 > 0$ such that

$$p_1 \cdot \eta = \delta, \qquad |\eta| \le C,$$

and

$$X \in \tilde{\Omega} \cap B_r, \quad 0 \le t \le s_0 \quad \Longrightarrow \quad X + t\eta \in \tilde{\Omega}. \tag{3.17}$$

Indeed, near $\partial\Omega$, a boundary chart admits n linearly independent inward unit directions. One of them has scalar product with p_1 of magnitude at least $c|p_1| \ge c\delta$, by (3.5). Reverse p_1 if necessary and rescale this inward vector to obtain η. The finite Lipschitz atlas gives uniform constants. Away from the boundary, take $\eta = \delta p_1/|p_1|^2$.

Fix $L > 1$ sufficiently large that

$$(1+L)^2 \frac{\lambda}{\Lambda} > 2^n \frac{\Lambda}{\lambda}, \tag{3.18}$$

and define

$$S(X) := X + \frac{L}{\delta}(\kappa + p_1 \cdot X)\eta.$$

Choose $\bar{q} \in A$ such that

$$\eta \cdot \bar{q} = \max_{q \in A} \eta \cdot q, \tag{3.19}$$

and set

$$\phi(X) := 2\tilde{u}(X) + \bar{q} \cdot (S(X) - X) - 2h - p \cdot X.$$

Since S is affine, ϕ is convex and $\mu_\phi = 2^n \mu_{\tilde{u}}$.

Claim. If $2L\kappa/\delta \le s_0$, then

$$S(D \cap \tilde{\Omega}) \subset \tilde{\Omega}, \tag{3.20}$$

$$\det DS = 1 + L, \tag{3.21}$$

$$\phi(X) = \tilde{u}(X) + \bar{q} \cdot (S(X) - X) \quad (X \in \partial D),$$

$$\tilde{u}(S(0)) - \phi(0) \ge h/2. \tag{3.22}$$

Proof of the claim. By (3.13),

$$0 < \frac{L}{\delta}(\kappa + p_1 \cdot X) < \frac{2L\kappa}{\delta} \qquad (X \in \overline{D}). \tag{3.23}$$

Since $D \subset B_r$, (3.20) follows from (3.17). Moreover,

$$\det DS = 1 + \frac{L}{\delta} p_1 \cdot \eta = 1 + L.$$

The boundary identity follows from $\tilde{u} = 2h + p \cdot X$ on ∂D. Finally, $\tilde{u}(0) = h$ gives $\phi(0) = \bar{q} \cdot S(0)$. Since $\tilde{v}(\bar{q}) \le -h/2$ by (3.15), duality yields

$$\tilde{u}(S(0)) \ge \bar{q} \cdot S(0) - \tilde{v}(\bar{q}) \ge \phi(0) + h/2,$$

proving the claim.

Step 4. A directional growth estimate. *Claim.* There exists $X_0 \in \partial D$ such that, with

$$t_0 := \frac{L}{\delta}(\kappa + p_1 \cdot X_0), \qquad S(X_0) = X_0 + t_0\eta,$$

we have $0 < t_0 < 2L\kappa/\delta$ and

$$\frac{\tilde{u}(X_0 + t_0\eta) - \tilde{u}(X_0)}{t_0} > \eta \cdot \bar{q} + \frac{r^2\delta}{16L}. \tag{3.24}$$

Proof of the claim. For every Borel set $F \subset D$,

$$\partial(\tilde{u} \circ S)(F) = (DS)^T \partial\tilde{u}(S(F)).$$

By (3.21), (3.20), (2.6) and (3.18), we obtain

$$\begin{aligned}\mu_{\tilde{u}\circ S}(F) &= (1+L)\mu_{\tilde{u}}(S(F))\\ &\geq (1+L)^2\frac{\lambda}{\Lambda}|F\cap S^{-1}(\tilde{\Omega})|\\ &\geq (1+L)^2\frac{\lambda}{\Lambda}|F\cap \tilde{\Omega}|\\ &\geq 2^n\mu_{\tilde{u}}(F)=\mu_{\phi}(F).\end{aligned}$$

Choose $X_0\in\partial D$ maximizing $\tilde{u}\circ S-\phi$ on ∂D. By the comparison principle (2.8) $\Big($take $\psi(\cdot)=\tilde{u}(S(\cdot))-(\tilde{u}(S(X_0))-\phi(X_0))\Big)$ and (3.22) give

$$\tilde{u}(S(X_0))-\phi(X_0)\geq \tilde{u}(S(0))-\phi(0)\geq h/2. \tag{3.25}$$

For t_0 defined above, (3.23) gives

$$0<t_0<2L\kappa/\delta. \tag{3.26}$$

Since $S(X_0)=X_0+t_0\eta$ and $\phi(X_0)=\tilde{u}(X_0)+t_0\eta\cdot\bar{q}$, we obtain

$$\begin{aligned}\frac{\tilde{u}(X_0+t_0\eta)-\tilde{u}(X_0)}{t_0} &= \eta\cdot\bar{q}+\frac{\tilde{u}(S(X_0))-\phi(X_0)}{t_0}\\ &\geq \eta\cdot\bar{q}+\frac{h}{2t_0}>\eta\cdot\bar{q}+\frac{r^2\delta}{16L},\end{aligned}$$

where the last inequality uses (3.12) and (3.26). This proves the claim.

Step 5. Enlargement of A. Define

$$\mathcal{L}_+(q):=X_0\cdot q-\tilde{u}(X_0).$$

By duality,

$$\tilde{v}-\mathcal{L}_+\geq 0\quad\text{on }\overline{\Omega^*}. \tag{3.27}$$

The directional growth estimate will give a target point separated from A, and the short displacement $S(X_0)-X_0=t_0\eta$ will control the increase of $\tilde{v}-\mathcal{L}_+$ along the segment from $\bar{q}$ to this point. We will move only far enough along this segment to obtain a separation comparable to δ.

Claim. There exists $p_+\in\overline{\Omega^*}$ such that

$$\begin{aligned}\operatorname{dist}(p_+,A)&\geq\frac{r^2\delta}{32LC},\\ \sup_{\operatorname{conv}(A\cup\{p_+\})}(\tilde{v}-\mathcal{L}_+)&\leq 2(n+1)\kappa.\end{aligned}$$

Proof of the claim. For $q\in A$, we have $-q/n\in A$. Similar to (3.14), by duality, (3.15) and (3.13), we have

$$\begin{aligned}(\tilde{v}-\mathcal{L}_+)(q)&\leq\tilde{v}(q)+n\tilde{v}(-q/n)+(n+1)\tilde{u}(X_0)\\ &\leq(n+1)\kappa\qquad(q\in A).\end{aligned} \tag{3.28}$$

Choose $q_*\in\overline{\Omega^*}$ such that

$$\tilde{u}(S(X_0))+\tilde{v}(q_*)=S(X_0)\cdot q_*.$$

Comparing this with $\tilde{u}(X_0)+\tilde{v}(q_*)\geq X_0\cdot q_*$, and using (3.24), we obtain

$$\eta\cdot(q_*-\bar{q})\geq\frac{\tilde{u}(S(X_0))-\tilde{u}(X_0)}{t_0}-\eta\cdot\bar{q}>\frac{r^2\delta}{16L}. \tag{3.29}$$

By duality, we have $\tilde{v}(\bar{q}) \geq S(X_0)\cdot\bar{q} - \tilde{u}(S(X_0))$, and then it follows

$$\begin{aligned}(\tilde{v}-\mathcal{L}_+)(q_*) - (\tilde{v}-\mathcal{L}_+)(\bar{q}) &= \tilde{v}(q_*) - \tilde{v}(\bar{q}) - X_0\cdot(q_*-\bar{q})\\ &\leq (S(X_0)-X_0)\cdot(q_*-\bar{q})\\ &= t_0\eta\cdot(q_*-\bar{q}).\end{aligned} \tag{3.30}$$

Thus the small factor t_0 controls the increase of the affine error along the target segment.

Set

$$\theta := \frac{r^2\delta}{32L\eta\cdot(q_*-\bar{q})}, \qquad p_+ := (1-\theta)\bar{q} + \theta q_*.$$

By (3.29), $0<\theta<1/2$, so $p_+\in\overline{\Omega^*}$. The choice of θ makes

$$\eta\cdot(p_+-\bar{q}) = \frac{r^2\delta}{32L}.$$

Since $\eta\cdot q\leq\eta\cdot\bar{q}$ for every $q\in A$, this gives

$$\operatorname{dist}(p_+,A) \geq \frac{\eta\cdot(p_+-\bar{q})}{|\eta|} \geq \frac{r^2\delta}{32LC}.$$

By convexity, (3.30), (3.28), and $t_0<2L\kappa/\delta$,

$$\begin{aligned}(\tilde{v}-\mathcal{L}_+)(p_+) &\leq (\tilde{v}-\mathcal{L}_+)(\bar{q}) + \theta t_0\eta\cdot(q_*-\bar{q})\\ &\leq (n+1)\kappa + \frac{r^2\delta}{32L}t_0\\ &\leq (n+1)\kappa + \frac{r^2\kappa}{16} \leq 2(n+1)\kappa.\end{aligned}$$

By the above estimate, (3.28) and convexity, we have $(\tilde{v}-\mathcal{L}_+)(q)\leq 2(n+1)\kappa$ for every $q\in\operatorname{conv}(A\cup\{p_+\})$ as claimed.

Finally, the propostion is proved by taking $c_* := \frac{r^2}{64LC}$, $K_* := 2(n+1)C_0$, and choosing $\varepsilon_*>0$ such that

$$\varepsilon_* \leq \min\left\{\frac{1}{2R_0C_0\operatorname{diam}\Omega^*}, \frac{s_0}{2LC_0}\right\}.$$

Since $\kappa\leq C_0\varepsilon$ and $\varepsilon\leq\varepsilon_*\delta\leq\varepsilon_*\operatorname{diam}\Omega^*$, this ensures $\kappa R_0\leq 1/2$ and $2L\kappa/\delta\leq s_0$. □

3.3. **A Hölder source boundary.** In this section we show that assuming $\partial\Omega$ to be Hölder source boundary is not enough for the global continuity of the optimal map. The following example is obtained from Pogorelov's solution [25], whose construction is recalled in [23, Section 4.4].

Proposition 3.3. *For every $n\geq 3$, let $\beta = \frac{n-2}{2(n-1)}$. There exist a bounded connected domain $\Omega\subset\mathbb{R}^n$ with $C^{0,\beta}$ boundary and a bounded convex domain $\Omega^*\subset\mathbb{R}^n$, with $|\Omega| = |\Omega^*|$, such that the quadratic optimal transport map from $\chi_\Omega\,dx$ to $\chi_{\Omega^*}\,dy$ has no continuous extension to $\overline{\Omega}$.*

Proof. Following the classical Pogorelov construction in [23, Section 4.4], denote $y = (y', y_n)\in\mathbb{R}^{n-1}\times\mathbb{R}$ and construct a solution of the form

$$v(y) = r^a Z(y_n), \qquad r = |y'|,$$

where $a = 2-2/n$, and $Z>0$ is to be chosen. To have $\det D^2 v = 1$, we solve the ODE

$$a^{n-1}Z^{n-2}\Big((a-1)ZZ'' - a(Z')^2\Big) = 1, \qquad Z(0)=1, \quad Z'(0)=0. \tag{3.31}$$

Standard ODE theory ensures that there exist $\tau_0 > 0$ and a positive even function $Z \in C^\infty((-2\tau_0, 2\tau_0))$ satisfying (3.31). Moreover, $D^2v(y) > 0$ for $y' \neq 0$. Since $a > 1$, v extends as a convex C^1 function across the axis $y' = 0$, where $v = Dv = 0$.

Let

$$\Omega^* := \{y \in \mathbb{R}^n : y_1 > 0,\ |y'| < 1,\ |y_n| < \tau_0\}, \qquad \Omega := Dv(\Omega^*).$$

The domain Ω^* is bounded and convex. Since $D^2v > 0$ there, Dv is a smooth diffeomorphism from Ω^* onto the bounded connected open set Ω. Define

$$u(x) := \max_{y \in \overline{\Omega^*}} \{x \cdot y - v(y)\}.$$

Then

$$Du = (Dv)^{-1} \quad \text{in } \Omega, \qquad (Du)_\#(\chi_\Omega\, dx) = \chi_{\Omega^*}\, dy.$$

The transport identity and $|\Omega| = |\Omega^*|$ follow from $\det D^2v = 1$.

We next determine the source domain near the origin. Let $a/(a-1) = 1/\beta$. For $x = Dv(y)$,

$$x' = aZ(y_n)r^{a-2}y', \qquad x_n = Z'(y_n)r^a, \qquad |x'| = aZ(y_n)r^{a-1}.$$

Hence x' has the same direction as y', and

$$\frac{x_n}{|x'|^{1/\beta}} = F(y_n), \qquad F(t) := \frac{Z'(t)}{(aZ(t))^{1/\beta}}.$$

The function F is odd, and (3.31) gives

$$F'(t) = \frac{(a-1)Z(t)Z''(t) - a(Z'(t))^2}{(a-1)a^{1/\beta}Z(t)^{1/\beta+1}} > 0.$$

Hence, F maps $(-\tau_0, \tau_0)$ bijectively onto $(-F(\tau_0), F(\tau_0))$.

We claim that, in a neighbourhood of the origin,

$$x \in \Omega \quad \Longleftrightarrow \quad x_1 > 0, \qquad |x_n| < F(\tau_0)|x'|^{1/\beta}. \tag{3.32}$$

If $x = Dv(y)$ with $y \in \Omega^*$, then $y_1 > 0$ implies $x_1 > 0$, and $y_n \in (-\tau_0, \tau_0)$ gives $|x_n| < F(\tau_0)|x'|^{1/\beta}$. Conversely, suppose that $x_1 > 0$ and $|x_n| < F(\tau_0)|x'|^{1/\beta}$. The first condition ensures $x' \neq 0$, while the second determines a unique $t \in (-\tau_0, \tau_0)$ such that $F(t) = \frac{x_n}{|x'|^{1/\beta}}$. Define

$$r := \left(\frac{|x'|}{aZ(t)}\right)^{1/(a-1)}, \qquad y' := r\frac{x'}{|x'|}, \qquad y_n := t.$$

Then $y_1 > 0$. Since $Z \geq 1$, we also have $r < 1$ whenever $|x'| < a$. By a direct computation we have $Dv(y) = x$. This proves (3.32) and shows that the restriction $|y'| < 1$ imposes no further condition near the origin.

To write the boundary as a graph, let $x'' = (x_2, \dots, x_{n-1})$ and the second inequality in (3.32) is equivalent to

$$x_1^2 + |x''|^2 > F(\tau_0)^{-2\beta}|x_n|^{2\beta}.$$

Together with $x_1 > 0$, this is equivalent to

$$x_1 > \rho(x'', x_n), \qquad \rho(x'', x_n) := \Big(F(\tau_0)^{-2\beta}|x_n|^{2\beta} - |x''|^2\Big)_+^{1/2}.$$

Hence $\partial\Omega$ near 0 is precisely the graph $x_1 = \rho(x'', x_n)$. Using $|\sqrt{s_+} - \sqrt{t_+}| \leq |s-t|^{1/2}$ and $0 < 2\beta < 1$, we obtain, on a bounded neighbourhood,

$$|\rho(x'', x_n) - \rho(z'', z_n)| \leq C\Big(|x_n - z_n|^\beta + |x'' - z''|^{1/2}\Big).$$

Since $\beta < 1/2$, this proves that $\rho \in C^{0,\beta}$.

Hence the boundary is $C^{0,\beta}$ at 0. The map Dv sends the axis $y' = 0$ to 0 and is injective on its complement in $\overline{\Omega^*}$, since v is strict convex along every segment not contained in the axis. At every other boundary point, Dv is a smooth local diffeomorphism at the unique corresponding point of $\partial\Omega^*$, and the source boundary is locally Lipschitz. Consequently Ω is a bounded $C^{0,\beta}$ domain.

Finally, for each fixed $t \in (-\tau_0, \tau_0)$, as $r \to 0$ we have

$$Dv(r, 0, \dots, 0, t) \to 0, \qquad Du(Dv(r, 0, \dots, 0, t)) = (r, 0, \dots, 0, t) \to (0, \dots, 0, t).$$

By taking different t, we have different limit of Du, which implies that Du has no continuous extension at $0 \in \partial\Omega$. □

4. Global Sobolev regularity

We prove Theorem 1.3 using the following proposition. Throughout this section, $|D^2\phi|$ denotes the operator norm.

Proposition 4.1. *Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be a bounded Lipschitz domain, and let $\phi : \mathbb{R}^n \to \mathbb{R}$ be globally Lipschitz and convex. Suppose that ϕ is strictly convex in Ω, differentiable on $\overline{\Omega}$, and satisfies*

$$\lambda\, dx \leq \mu_\phi \leq \Lambda\, dx \qquad \text{in } \Omega, \tag{4.1}$$

and

$$|D\phi(x) - D\phi(y)| \leq K|x - y|^\alpha \qquad (x, y \in \overline{\Omega}), \tag{4.2}$$

where $0 < \lambda \leq \Lambda < \infty$, $\alpha \in (0, 1)$ and $K < \infty$. Then $\phi \in W^{2,1+\varepsilon}(\Omega)$ for some $\varepsilon = \varepsilon(n, \lambda, \Lambda, \alpha) > 0$, and

$$\int_\Omega |D^2\phi|^{1+\varepsilon}\, dx \leq C,$$

where C depends only on $n, \lambda, \Lambda, \alpha, K$, $\mathrm{Lip}(\phi)$, and the diameter and Lipschitz character of Ω.

We first establish a local estimate with polynomial dependence on the gradient bound. This dependence will allow us to apply the estimate on balls approaching $\partial\Omega$.

Lemma 4.2. *Let $\varphi \in C^{1,\alpha}(B_2)$ be strictly convex, with*

$$\lambda\, dx \leq \mu_\varphi \leq \Lambda\, dx \quad \text{in } B_2, \qquad 1 + \|D\varphi\|_{L^\infty(B_2)} + [D\varphi]_{C^\alpha(B_2)} \leq K.$$

There are $\delta > 0$, $N \geq 1$ and $C < \infty$, depending only on $n, \lambda, \Lambda, \alpha$, such that

$$\int_{B_{1/2}} (\Delta\varphi)^{1+\delta}\, dx \leq CK^N. \tag{4.3}$$

Proof. We combine the normalized interior estimate of De Philippis–Figalli–Savin [15, Theorem 1.1] with the section arguments in the proofs of [22, Lemmas 3.4–3.5]. The $C^{1,\alpha}$ bound allows us to control the dependence on K by a polynomial. Throughout the proof, C and N depend only on $n, \lambda, \Lambda, \alpha$ and may increase from line to line.

We group points according to the height of their largest section contained in B_1. For $x \in B_{1/2}$, denote

$$S_h(x) := \{y \in B_2 : \varphi(y) < \varphi(x) + D\varphi(x) \cdot (y - x) + h\},$$

and set

$$\bar{h}(x) := \min_{y \in \partial B_1} \big(\varphi(y) - \varphi(x) - D\varphi(x) \cdot (y - x)\big).$$

By strict convexity of φ, we have $\bar{h}(x) > 0$. By the definition we have that $S_{\bar{h}(x)}(x) \subset B_1$. By (2.9) we have $\bar{h}(x) \leq C$.

Step 1. Vitali covering. For $\gamma > 0$, let

$$F_\gamma := \{x \in B_{1/2} : \gamma/2 \le \bar{h}(x) < \gamma\}.$$

We first choose a vitali type covering. By [22, Lemma 2.5], there exist $x_i \in F_\gamma$, $h_i = \bar{h}(x_i) \in [\gamma/2, \gamma)$, and a fixed $\vartheta = \vartheta(n, \lambda, \Lambda) \in (0, 1)$ such that

$$F_\gamma \subset \bigcup_{i=1}^{M_\gamma} S_{h_i/2}(x_i), \qquad S_{\vartheta h_i}(x_i) \text{ are pairwise disjoint.} \tag{4.4}$$

We choose ϑ small enough that, by [22, Lemma 2.5(1)],

$$2S_{\vartheta^2 h_i}(x_i) \subset S_{\vartheta h_i}(x_i) \subset B_1.$$

Here $2S_h(x_i)$ denotes the dilation of $S_h(x_i)$ by a factor of two about x_i.

We next use the local mass estimate established in the proof of [22, Lemma 3.5]. Since $\overline{S_{h_i}(x_i)}$ touches ∂B_1 and $x_i \in B_{1/2}$, this section has diameter at least $1/2$. By convexity and the section balance estimate [22, Lemma 2.4], $S_{\vartheta^2 h_i}(x_i)$ contains a segment centred at x_i of uniformly positive length. For $w(x) := \varphi(x) + \frac{1}{2}|x|^2$, by the argument at the end of the proof of [22, Lemma 3.5], we have

$$\mu_w\Big(2S_{\vartheta^2 h_i}(x_i)\Big) \ge c\gamma^{n/2-1}, \tag{4.5}$$

where $c > 0$ depends only on n, λ, Λ.

The sets $2S_{\vartheta^2 h_i}(x_i) \subset B_1$ are pairwise disjoint. Summing (4.5) yields

$$cM_\gamma \gamma^{n/2-1} \le \sum_{i=1}^{M_\gamma} \mu_w\Big(2S_{\vartheta^2 h_i}(x_i)\Big) \le \mu_w(B_1).$$

Finally, since $Dw(x) = D\varphi(x) + x$ and $|D\varphi| \le K$, we have $\mu_w(B_1) \le CK^n$. Hence,

$$M_\gamma \gamma^{n/2-1} \le CK^n. \tag{4.6}$$

Step 2. Hessian estimates on sections. For each fixed i, we subtract an affine function so that

$$\varphi = 0 \quad \text{on } \partial S_{h_i}(x_i), \qquad \min_{S_{h_i}(x_i)} \varphi = \varphi(x_i) = -h_i.$$

By John's lemma [21], there exists a point $z_i \in S_{h_i}(x_i)$ and a positive definite matrix A_i, such that

$$B_1 \subset U_i := A_i^{-1}(S_{h_i}(x_i) - z_i) \subset B_n.$$

Let d_i is the smallest eigenvalue of A_i. Consider

$$\tilde{\varphi}_i(y) := (\det A_i)^{-2/n} \varphi(z_i + A_i y).$$

Then, $\tilde{\varphi}_i = 0$ on ∂U_i. This rescaling preserves the Monge–Ampère measure bounds. Moreover, by (2.9) we have

$$ch_i^{n/2} \le \det A_i \le Ch_i^{n/2}, \qquad \|A_i^{-1}\| = d_i^{-1}.$$

It straightforward to verify that

$$U_i' := A_i^{-1}(S_{h_i/2}(x_i) - z_i) = \left\{\tilde{\varphi}_i < \frac{1}{2} \min_{U_i} \tilde{\varphi}_i\right\}.$$

By [15, Theorem 1.1], we have

$$\int_{U_i'} (\Delta \tilde{\varphi}_i)^{1+\delta_0}\, dy \le C$$

for some $\delta_0 = \delta_0(n, \lambda, \Lambda) > 0$.

To return to the original variables, write $x = z_i + A_i y$. The Hessians satisfy

$$D^2\varphi(x) = (\det A_i)^{2/n} A_i^{-T} D^2\tilde{\varphi}_i(y) A_i^{-1}.$$

Since $D^2\tilde{\varphi}_i \geq 0$, it follows that

$$\Delta\varphi(x) \leq (\det A_i)^{2/n} \|A_i^{-1}\|^2 \Delta\tilde{\varphi}_i(y) \leq C\frac{h_i}{d_i^2}\Delta\tilde{\varphi}_i(y).$$

Using $dx = (\det A_i)\, dy$, we conclude that

$$\begin{aligned} \int_{S_{h_i/2}(x_i)} (\Delta\varphi)^{1+\delta_0}\, dx &\leq C(\det A_i)\left(\frac{h_i}{d_i^2}\right)^{1+\delta_0} \int_{U_i'} (\Delta\tilde{\varphi}_i)^{1+\delta_0}\, dy \\ &\leq C h_i^{n/2}\left(\frac{h_i}{d_i^2}\right)^{1+\delta_0}. \end{aligned} \tag{4.7}$$

Step 3. Hessian estimates on $B_{1/2}$. We next fix $\delta := \min\{\delta_0, \alpha\}$. By Hölder's inequality and (2.9), the same bound holds with δ_0 replaced by δ. Since $[D\varphi]_{C^\alpha(B_2)} \leq K$, we have

$$\varphi(y) - \varphi(x_i) - D\varphi(x_i)\cdot(y - x_i) \leq K|y - x_i|^{1+\alpha}.$$

Consequently, $S_{h_i}(x_i)$ contains a ball centred at x_i of radius $(h_i/K)^{1/(1+\alpha)}$, and hence

$$d_i \geq c(h_i/K)^{1/(1+\alpha)} \tag{4.8}$$

for some universal constant c. Recall the definition of F_γ in Step 1, and notice that $\frac{1}{2}\gamma \leq h_i < \gamma$.

By (4.7), (4.8), (4.6) and summing over the covering in (4.4), we obtain

$$\begin{aligned} \int_{F_\gamma} (\Delta\varphi)^{1+\delta}\, dx &\leq C M_\gamma \gamma^{n/2}\left(K^{2/(1+\alpha)}\gamma^{-(1-\alpha)/(1+\alpha)}\right)^{1+\delta} \\ &\leq C K^N \gamma^{\frac{2\alpha-(1-\alpha)\delta}{1+\alpha}}. \end{aligned}$$

Since $\delta \leq \alpha$, the exponent of γ is at least $\alpha > 0$. Finally, $0 < \bar{h}(x) \leq C$ on $B_{1/2}$, so summing over dyadic values of $\gamma = 2^{-j}(\sup_{B_{1/2}} \bar{h} + 1)$ yields

$$\int_{B_{1/2}} (\Delta\varphi)^{1+\delta}\, dx \leq C K^N \sum_{j=0}^{\infty} 2^{-j\alpha} \leq C K^N.$$

□

Proof of Proposition 4.1. We combine integral estimates for the Hessian near and away from the boundary.

Step 1. Hessian estimates near the boundary. By [15, Theorem 1.1], the distributional Hessian $D^2\phi$ is absolutely continuous with respect to Lebesgue measure in Ω. Choose a nonnegative $\zeta \in C_c^\infty(\mathbb{R}^n)$ such that $\zeta = 1$ on $\overline{\Omega}$ and $\int_{\mathbb{R}^n} |D\zeta|\, dx \leq C$, where C depends only on n and $\operatorname{diam}\Omega$. Since ϕ is convex, we have

$$\int_\Omega \Delta\phi\, dx \leq \int_{\mathbb{R}^n} \zeta\, d(\Delta\phi) = -\int_{\mathbb{R}^n} D\phi\cdot D\zeta\, dx \leq \operatorname{Lip}(\phi)\int_{\mathbb{R}^n} |D\zeta|\, dx \leq C.$$

Since $|D^2\phi| \leq \Delta\phi$ by convexity, this bound implies $\phi \in W^{2,1}(\Omega)$. Set

$$d(x) = \operatorname{dist}(x, \partial\Omega) \quad (x \in \Omega).$$

Since Ω is a bounded Lipschitz domain, a family of at most Cr^{1-n} balls $B_{2r}(x_j)$, with centres $x_j \in \partial\Omega$, cover $\{x \in \Omega : d(x) < r\}$ for all sufficiently small $r > 0$. Let $\zeta_j \in C_c^\infty(B_{4r}(x_j))$ be cut off functions such that

$$0 \le \zeta_j \le 1, \qquad \zeta_j = 1 \text{ on } B_{2r}(x_j), \qquad |D\zeta_j| \le C/r. \tag{4.9}$$

Let $p_j := D\phi(x_j)$. Denote by ν the outward unit normal to $\partial\Omega$, defined $\mathcal{H}^{n-1}$-almost everywhere on $\partial\Omega$. By (4.2), $D\phi$ is continuous on $\overline{\Omega}$, so its Sobolev trace coincides with $D\phi|_{\partial\Omega}$. Since $D\phi \in W^{1,1}(\Omega)$, the Gauss-Green formula implies

$$\int_\Omega \zeta_j \Delta\phi \, dx = \int_{\partial\Omega} \zeta_j (D\phi - p_j)\cdot \nu \, d\mathcal{H}^{n-1} - \int_\Omega (D\phi - p_j)\cdot D\zeta_j \, dx. \tag{4.10}$$

By (4.2), we see that

$$|D\phi - p_j| \le CKr^\alpha \quad \text{on} \quad supp\, \zeta_j \cap \overline{\Omega}.$$

This together with (4.9) shows that both terms on the right of (4.10) are bounded in absolute value by $Cr^{n-1+\alpha}$. Summing over the cover yields

$$\int_{\{d(x)<r\}} \Delta\phi \, dx \le Cr^\alpha. \tag{4.11}$$

Step 2. Hessian estimates away from the boundary. For x such that $B_{2s}(x) \subset \Omega$, let

$$\varphi_{x,s}(y) = \frac{\phi(x+sy) - \phi(x) - sD\phi(x)\cdot y}{s^2}, \qquad y \in B_2.$$

For every Borel set $F \subset B_2$, we have by change of variables,

$$\mu_{\varphi_{x,s}}(F) = s^{-n}\mu_\phi(x+sF).$$

Since $|x+sF| = s^n|F|$, the measure bounds (4.1) imply

$$\lambda|F| \le \mu_{\varphi_{x,s}}(F) \le \Lambda|F|.$$

Thus the Monge–Ampère bounds are unchanged. Moreover,

$$1 + \|D\varphi_{x,s}\|_{L^\infty(B_2)} + [D\varphi_{x,s}]_{C^\alpha(B_2)} \le C(1+K)s^{\alpha-1} \qquad (0 < s \le 1).$$

Applying Lemma 4.2 to $\varphi_{x,s}$, we obtain

$$\int_{B_{1/2}} (\Delta\varphi_{x,s})^{1+\delta} \, dy \le C\Big[C(1+K)s^{\alpha-1}\Big]^N \le Cs^{-N(1-\alpha)}.$$

The change of variables $z = x + sy$ then shows $\Delta\varphi_{x,s}(y) = \Delta\phi(z)$. Hence

$$\int_{B_{s/2}(x)} (\Delta\phi)^{1+\delta} \, dz = s^n \int_{B_{1/2}} (\Delta\varphi_{x,s})^{1+\delta} \, dy \le Cs^{n-N(1-\alpha)}.$$

Taking $s = r/4$ and covering $\{d \ge r\}$ by at most Cr^{-n} such balls, we obtain

$$\int_{\{d\ge r\}} (\Delta\phi)^{1+\delta} \, dx \le Cr^{-N(1-\alpha)}. \tag{4.12}$$

Step 3. Global integrability. By (4.11) and (4.12),

$$\begin{aligned}\int_{\{\Delta\phi>t\}} \Delta\phi \, dx &\le \int_{\{d<r\}} \Delta\phi \, dx + t^{-\delta}\int_{\{d\ge r\}} (\Delta\phi)^{1+\delta} \, dx \\ &\le C\Big(r^\alpha + t^{-\delta} r^{-N(1-\alpha)}\Big).\end{aligned}$$

For sufficiently large t, choose $r = t^{-\delta/(\alpha+N(1-\alpha))}$. Then we have

$$\int_{\{\Delta\phi>t\}} \Delta\phi \, dx \le Ct^{-\tau}, \quad \text{where } \tau = \frac{\alpha\delta}{\alpha + N(1-\alpha)} > 0. \tag{4.13}$$

Set $\varepsilon := \tau/2$ and take $T \geq 1$ sufficiently large. We deduce by (4.13) that

$$\begin{aligned}\int_\Omega (\Delta\phi)^{1+\varepsilon}\,dx &= \varepsilon \int_0^\infty t^{\varepsilon-1} \int_{\{\Delta\phi > t\}} \Delta\phi\,dx\,dt \\ &\leq C\varepsilon \left(\int_0^T t^{\varepsilon-1}\,dt + \int_T^\infty t^{\varepsilon-\tau-1}\,dt \right) \leq C.\end{aligned}$$

Since $|D^2\phi| \leq \Delta\phi$ by convexity, the proposition follows. □

Proof of Theorem 1.3. The minimal convex extension of u is globally Lipschitz, strictly convex in Ω, and satisfies (2.6). Theorem 1.1 implies (4.2) for some $\alpha \in (0,1)$. We apply Proposition 4.1 with measure bounds λ/Λ and Λ/λ. After subtracting an affine function, $\mathrm{Lip}(u) \leq \operatorname{diam}\Omega^*$, so all constants in (1.4) depend only on the data in Theorem 1.1. □

5. Optimal partial transport

In this section, we assume the domains, densities and mass satisfy the assumptions of Corollary 1.4. Adding the inactive mass gives the two completed transports [7, Lemma 7.2]:

$$\begin{aligned}(Du)_\#\Big[(f\chi_U + g\chi_{\Omega^*\setminus V})\,dx\Big] &= g\chi_{\Omega^*}\,dy, \\ (Dv)_\#\Big[(g\chi_V + f\chi_{\Omega\setminus U})\,dy\Big] &= f\chi_\Omega\,dx.\end{aligned} \tag{5.1}$$

We extend u and v to $\mathbb{R}^n$ as in (2.2), using their respective convex targets Ω^* and Ω, and denote their Legendre transforms by u^* and v^*. By [7, Theorem 6.3], $u, v \in C^1(\mathbb{R}^n)$ and $Du : \overline{U} \to \overline{V}$ is a homeomorphism with inverse Dv. We choose their additive constants so that

$$\begin{aligned}U &= \Omega \cap \{x : u(x) > |x|^2/2\}, \\ V &= \Omega^* \cap \{y : v(y) > |y|^2/2\}.\end{aligned} \tag{5.2}$$

On the inactive parts,

$$Du(z) = z \quad (z \in \overline{\Omega^* \setminus V}), \qquad Dv(x) = x \quad (x \in \overline{\Omega \setminus U}).$$

By the quadratic obstacle relation in [7, Corollary 4.5 and Theorem 6.3], there is a constant $a_0 \in \mathbb{R}$, common to all inactive components, such that

$$\begin{aligned}u^*(y) &> \frac{1}{2}|y|^2 + a_0 \quad (y \in V), \\ u^*(y) &= \frac{1}{2}|y|^2 + a_0 \quad (y \in \overline{\Omega^* \setminus V}).\end{aligned} \tag{5.3}$$

In particular, $u^*(y) \geq |y|^2/2 + a_0$ on $\overline{\Omega^*}$. An analogous statement holds for v^* with the two measures interchanged. We also use the interior ball condition [7, Lemma 6.8]:

$$\Omega \cap B_{|x-y|}(y) \subset U, \qquad \Omega^* \cap B_{|x-y|}(x) \subset V, \qquad x \in \overline{U}, \quad y = Du(x). \tag{5.4}$$

The free boundary in the source domain is defined as $\mathcal{F} := \partial U \cap \Omega$. The unit inner normal of free boundary is given by [7, Remark 6.5]:

$$\nu_{\mathcal{F}}(x) = \frac{Du(x) - x}{|Du(x) - x|}. \tag{5.5}$$

The next lemma combines the proof of [12, Theorem 1.1] with the exclusion of pairs of nontransverse intersection points in [20, Proposition 4.4]. In contrast to the case treated in [12, Theorem 1.1], here we allow both measures to have inactive parts.

Lemma 5.1. *The active regions U and V are bounded Lipschitz domains.*

Proof. By (5.2) and [7, Theorem 6.3 and Remark 6.5], the free boundary is locally the zero level set of the C^1 function $x \mapsto u(x) - |x|^2/2$. Its gradient $Du(x) - x$ is nonzero on $\overline{\mathcal{F}}$ and gives the inward normal direction. Hence $\overline{\mathcal{F}}$ is C^1. To show U is a Lipschitz domain, we only need to exclude $x_0 \in \overline{\mathcal{F}} \cap \partial\Omega$ such that

$$(Du(x_0) - x_0) \cdot (x - x_0) \le 0 \qquad (x \in \Omega). \tag{5.6}$$

Suppose that (5.6) holds, and set $y_0 = Du(x_0)$. By [20, Proposition 4.4], y_0 is not a nontransverse intersection point of the target free and fixed boundaries. Hence V is a Lipschitz domain in a neighbourhood of y_0. Consequently,

$$|V \cap B_r(y_0)| \ge C^{-1} r^n \qquad \text{for all sufficiently small } r > 0. \tag{5.7}$$

We now use the argument in [12, proof of Theorem 1.1, equations (3.1)–(3.5)]. Translate x_0 to zero and rotate so that $y_0 = de_n$, where $d > 0$. Then $\Omega \subset \{x_n < 0\}$. Put $G_r = V \cap B_r(y_0)$. For $p \in G_r$ and $x = Dv(p)$, the ball condition (5.4) implies $|p - x| \le |p|$. Otherwise a relative neighbourhood of zero in Ω would lie in U, contrary to $0 \in \overline{\mathcal{F}}$. By [12, equation (3.3)–(3.4)] we have

$$Dv(G_r) \subset \{x : |x'| \le 2r,\ -Cr^2 \le x_n < 0\}.$$

By (5.1) and (1.1) we have

$$|G_r| \le C|Dv(G_r)| \le Cr^{n+1},$$

contradicting (5.7).

Thus (5.6) cannot occur. By the local graph representations and compactness we have that ∂U is Lipschitz. Connectedness follows from [7, Corollary 6.7]. Interchanging the two measures proves the assertion for V. □

Proof of Corollary 1.4. By Lemma 5.1, U and V are Lipschitz. The first transport in (5.1) has convex target Ω^*. To apply the proof of Theorem 1.1, we use the obstacle relation to keep the localization point in $\overline{U}$. Indeed, with $B := \Omega^* \setminus V$, (5.1) gives

$$C^{-1}\chi_U \, dx \le \mu_u \le C\chi_{U \cup B} \, dx. \tag{5.8}$$

Take r in Proposition 3.1 sufficiently small that $r < \operatorname{dist}(\overline{\Omega}, \overline{\Omega^*})/4$.

We first establish the following analogue of Proposition 3.1 for u^*.

Claim. There exist constants $\varepsilon_0, c_* > 0$ and $K \ge 2$ with the following property. Let $A \subset \overline{\Omega^*}$ be compact and convex, with $\delta = \operatorname{diam} A > 0$, and let $\mathcal{L}$ be affine with $\mathcal{L} \le u^*$ on $\overline{\Omega^*}$. If

$$\sup_A (u^* - \mathcal{L}) \le \varepsilon, \qquad 0 < \varepsilon \le \varepsilon_0 \delta^2,$$

then there exist $p_+ \in \overline{\Omega^*}$ and an affine function $\mathcal{L}_+ \le u^*$ on $\overline{\Omega^*}$ such that

$$\operatorname{dist}(p_+, A) \ge c_* \delta, \qquad \sup_{\operatorname{conv}(A \cup \{p_+\})} (u^* - \mathcal{L}_+) \le K\varepsilon.$$

Apply the proof of Proposition 3.1 with u^* as the dual potential. For $A, \mathcal{L}, \delta, \varepsilon$ as in that proposition, assume

$$0 < \varepsilon \le \varepsilon_0 \delta^2, \tag{5.9}$$

where $\varepsilon_0 > 0$ is small enough that $\varepsilon_0 \operatorname{diam} \Omega^* \le \varepsilon_*$. By the argument in Step 1 of the proof of Proposition 3.1 leading to (3.11), applied to the completed transport (5.1), we have $x_0 \in \overline{U} \cup \overline{B}$. Combining (3.14) in Step 2 with (3.8), and undoing the initial paired affine normalization, yields

$$0 \le u(x_0) + u^*(q) - x_0 \cdot q \le C\varepsilon \qquad (q \in A). \tag{5.10}$$

If $x_0 \in \overline{B}$, then $Du(x_0) = x_0$. By Legendre duality and (5.3) we have

$$\begin{aligned} u(x_0) + u^*(q) - x_0 \cdot q &= u^*(q) - u^*(x_0) + |x_0|^2 - x_0 \cdot q \\ &\geq \Big(\frac{1}{2}|q|^2 + a_0\Big) - \Big(\frac{1}{2}|x_0|^2 + a_0\Big) + |x_0|^2 - x_0 \cdot q \\ &= \frac{1}{2}|q - x_0|^2 \qquad (q \in A). \end{aligned}$$

Combining this with (5.10) we have $\delta^2 \leq C\varepsilon$, contradicting (5.9) when ε_0 is sufficiently small. Thus $x_0 \in \overline{U}$.

We now follow Steps 2–5 of the proof of Proposition 3.1. Write the section D from Step 2 in the original source variables. By (3.13), we have $\overline{D} \subset B_r(x_0)$. Since $x_0 \in \overline{U}$ and $r < \operatorname{dist}(\overline{\Omega}, \overline{\Omega^*})/4$, we have $\overline{D} \cap \overline{B} = \varnothing$. Thus (5.8) gives $\mu_u \leq C\chi_U \, dx$ in D.

In Step 3, by the Lipschitz regularity of U, we can choose the inward direction and translation length similar as in (3.17). Choose L as in (3.18), using the constants in (5.8). By (3.9) and (5.9),

$$\frac{2L\kappa}{\delta} \leq 2LC_0\varepsilon_0\delta \leq 2LC_0\varepsilon_0 \operatorname{diam}\Omega^* \leq s_0,$$

after decreasing ε_0. Hence the affine map S from Step 3, written in the original variables, satisfies

$$S(D \cap U) \subset U, \qquad \det DS = 1 + L,$$

as in (3.20) and (3.21).

The measure comparison in Step 4 therefore remains valid. Indeed, for every Borel set $F \subset D$,

$$\begin{aligned} \mu_{u\circ S}(F) &= (1 + L)\mu_u(S(F)) \\ &\geq C^{-1}(1 + L)^2|F \cap S^{-1}(U)| \\ &\geq C^{-1}(1 + L)^2|F \cap U| \\ &\geq 2^n\mu_u(F) = \mu_\phi(F), \end{aligned}$$

where ϕ is the corresponding comparison function constructed in Step 3. The comparison principle then gives (3.25) and (3.24), exactly as in Step 4. Finally, the duality and interpolation argument in Step 5, using (3.29) and (3.30), gives (3.1) for u^* under the quadratic threshold (5.9). This proves the claim.

We next repeat the iteration in Lemma 3.2, with continuation condition

$$K_*^j\varepsilon \leq \varepsilon_0\delta_j^2, \qquad \delta_j = \operatorname{diam} A_j.$$

The estimates (3.3) and (3.4) are unchanged. At the stopping index $j_* \leq \bar{N}$, we now have

$$K_*^{j_*}\varepsilon > \varepsilon_0\delta_{j_*}^2 \geq \varepsilon_0\delta^2.$$

Consequently, by (3.4) we have

$$\varepsilon > \varepsilon_0\delta^2 K_*^{-\bar{N}} \geq C^{-1}\delta^{2+N\log_2 K_*}.$$

Letting $\varepsilon \to \sup_A(u^* - \mathcal{L})$ yields

$$\sup_A(u^* - \mathcal{L}) \geq C^{-1}(\operatorname{diam} A)^s, \qquad s := 2 + N\log_2 K_* > 2. \tag{5.11}$$

By the duality argument in the proof of Theorem 1.1, with $\alpha = 1/(s-1)$, we have

$$|Du(x^{(0)}) - Du(x^{(1)}))| \leq C|x^{(0)} - x^{(1)}|^\alpha \qquad (x^{(0)}, x^{(1)} \in \mathbb{R}^n). \tag{5.12}$$

In particular, $u \in C^{1,\alpha}(\overline{U})$. By (5.5) we have $\overline{\mathcal{F}} \in C^{1,\alpha}$. By interchanging the two measures and decreasing α if necessary we have the same conclusion for $\overline{\mathcal{F}^*}$. □

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